Journal of Cardiothoracic and Vascular Anesthesia 39 (2025) 2031–2039

Contents lists available at ScienceDirect

Journal of Cardiothoracic and Vascular Anesthesia

journal homepage: www.jcvaonline.com

Original Article

Implementing Ultra–Fast-Track Cardiac Anesthesia in Minimally Invasive Cardiac Surgery

Tian Jiang, BS*, Linting Xu, BS*, Qinghui Zheng, MD†, Yihui Zhang, BS‡, Shuaibing Wang, BS‡, Yu Wang, MD*, Xiaokan Lou, BS*, Meijuan Yan, MD*, Hanwei Wei, MD*,[1]

*Center for Rehabilitation Medicine, Department of Anesthesiology, Zhejiang Provincial People's Hospital, Affiliated People's Hospital, Hangzhou Medical College, Hangzhou, Zhejiang, China

†General Surgery, Cancer Center, Department of Breast Surgery, Zhejiang Provincial People's Hospital, Affiliated People's Hospital, Hangzhou Medical College, Hangzhou, Zhejiang, China

‡Jinzhou Medical University, Jinzhou, Liaoning Province, China

Objective: To analyze factors influencing the implementation of ultra–fast-track cardiac anesthesia (UFTCA) from the perspectives of preoperative and intraoperative conditions and to compare its postoperative recovery with that of conventional cardiac anesthesia (CGA).

Design: An observational retrospective study

Setting: Electronic medical records data for patients from January 2021 through July 2023

Participants: 967 patients with a documented history of minimally invasive cardiac surgery (MICS)

Interventions: Review of electronic medical records

Measurements and Results: Data for 947 patients, including 439 who received UFTCA and 508 who received CGA, were analyzed retrospectively. Multivariate logistic regression identified independent factors associated with UFTCA implementation. Risk factor analysis showed that age >50 years (odds ratio [OR], 1.924; p = 0.040), history of stroke (OR, 2.290; p = 0.009), and duration of cardiopulmonary bypass (CPB) (OR, 1.013; p < 0.001), intraoperative sufentanil dosage (OR, 1.035; p < .001), and surgeries ending after 8 pm (OR, 2.184; p < 0.001) were negatively correlated with UFTCA implementation. Pectoral muscle fascial plane block (OR, 0.120; p < .001) and dexamethasone (OR, 0.438; p < .001) were positive factors in UFTCA implementation. Compared to the CGA group, the UFTCA group had fewer cases of intensive care unit (ICU) rescue analgesia (77 [17.5%] vs 178 [35%]; p < 0.001), shorter ICU stay (mean, 22.83 ± 20 hours vs 44 ± 43 hours; p < 0.001) and postoperative hospital stay (mean, 8 ± 3 days vs 10 ± 5 d; p < .001), and lower incidence of postoperative delirium (6 [1.4%] vs 28 [5.5%]; p = 0.001) but a higher incidence of postoperative nausea and vomiting (86 [19.6%] vs 62 [12.2%]; p = 0.002). The oxygenation index (OI) was lower at 1 hour postsurgery but higher at 6 hours postsurgery in the UTCA group compared to the CGA group (p < 0.05).

Conclusions: Seven independent factors were identified for UFTCA implementation in MICS, among which age >50 years, history of stroke, duration of CPB, intraoperative sufentanil dosage, and surgery ending after 8 pm were negative factors and pectoral muscle fascial plane block and dexamethasone use were positive factors. The use of UFTCA in MICS shortened ICU stay and hospital stay, decreased the incidence of delirium, and promoted postoperative pulmonary function.

Key Words: ultra–fast-track cardiac anesthesia; minimally invasive cardiac surgery; risk factors; enhanced recovery after cardiac surgery; cardiopulmonary bypass

This work was supported by Medical Science and Technology Project of Zhejiang Province (Grants 2022KY062, 2022KY037, 2024KY009, 2024KY657, and 2024KY779).

[1]*Address correspondence to* Han-Wei Wei, MD, Center for Rehabilitation Medicine, Department of Anesthesiology, Zhejiang Provincial People's Hospital, Affiliated People's Hospital, Hangzhou Medical College, 158 Shangtang Rd, Gongshu District, Hangzhou, Zhejiang, China.

E-mail address: omadrr@163.com (H. Wei).

https://doi.org/10.1053/j.jvca.2025.04.009

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MINIMALLY INVASIVE CARDIAC SURGERY (MICS) refers to heart surgeries performed through small incisions, including direct-vision surgery through mini-incisions, total endoscopic surgery, and robotic surgery. In terms of surgical approaches, small incisions can be made in the upper sternum, parasternum, or right anterolateral intercostal space. MICS offers a plethora of advantages over conventional open-chest procedures, including diminished trauma, minimized blood loss, and decreased postoperative complications.[1] Consequently, MICS has garnered widespread clinical acceptance.[2,3] At the authors' center, the MICS approach involves a small incision between the anterior axillary line and the mid-axillary line in the right underarm region. This approach, which is more discreet and aesthetical, has been used since 2018 and has a high success rate of approximately 83%.

Enhanced recovery after surgery (ERAS) endeavors to standardize perioperative care, aiming to mitigate complications, enhance the patient's experience, and optimize medical resource utilization.[4] However, research on enhanced recovery after cardiac surgery (ERACS) remains in the early stages. Fast-track cardiac anesthesia (FTCA) and ultra−fast-track cardiac anesthesia (UFTCA) have pivotal roles in implementing ERACS.[5] FTCA typically entails extubation within 6 to 8 hours after operation,[6,7] while UFTCA aims for extubation within 1 hour or immediately after operation.[8,9]

Despite ongoing debates and skepticism,[10,11] UFTCA has shown promise in hastening postoperative recovery and reducing healthcare costs.[7,12] Nevertheless, the research maturity of ERACS lags notably behind that of other surgical fields.[5] Widespread implementation of UFTCA is hindered by various clinical factors, including patient-related factors such as advanced age, preoperative comorbidities (eg, history of stroke, diabetes, obesity), intraoperative factors such as the duration of surgery and cardiopulmonary bypass (CPB), and dosage of anesthetic agents.[5,8]

Since 2021, the authors' institution has integrated UFTCA into MICS protocols to furnish substantive evidence for clinical application. In the early stages of implementing UFTCA at the center, admission criteria were established as American Society of Anesthesiologists class I to III, New York Heart Association (NYHA) cardiac function class I to III, and age 18 years or older.[13] However, as work progressed, surgeons and anesthesiologists adjusted these criteria, jointly determining whether to implement UFTCA based on the patient's preoperative and intraoperative conditions. The implementation and success of UFTCA may be influenced by such factors such as age, preoperative comorbidities, duration of surgery, and the dosage of anesthetic drugs used during the procedure. However, the main factors affecting UFTCA implementation and success remain unclear. Therefore, this retrospective study was conducted to identify the main factors influencing the implementation and success of UFTCA, with the aim of providing a reference for clinical decision making.

Patients and Methods

This retrospective study collated clinical data from relevant patients to explore independent influencing factors of UFTCA implementation in MICS and to compare the effects of UFTCA with CGA on patient recovery.

General Information

The medical records of 967 patients who underwent MICS via right lateral thoracotomy at the study center between January 2021 and July 2023 were reviewed retrospectively. This study was approved by the center's Ethical Review Committee (approval QT2024007), with a waiver of patient informed consent.

The patients were categorized into either the UFTCA group or the CGA group based on the anesthesia modality used. Patients undergoing MICS who were extubated within 1 hour or immediately after operation were assigned to the UFTCA group, and the remaining patients composed the CGA group, which included patients who were planned for CGA, those switched from UFTCA, and those who failed UFTCA. Exclusion criteria included cardiogenic shock, septic shock, extracorporeal membrane oxygenation (ECMO) support, left ventricular assist device (LVAD) implantation, and significant missing data. All clinical data were retrieved from the hospital's electronic medical records system.

Anesthesia Procedures

During the study period, the UFTCA group followed a more standardized perioperative management protocol, while the approach for the CGA group depended more on the anesthesiologist's experience and preference. All anesthesia was administered by experienced cardiac anesthesiologists, defined as those who received training in cardiac anesthesia and are capable of independently managing anesthesia for various cardiac surgeries, as well as proficiently assessing and handling diverse cardiac pathologic conditions.

Patients received preoperative education before entering the operating room. On entry, routine monitoring, invasive arterial blood pressure and central venous catheterization were performed. Anesthesia was induced using etomidate, propofol, sufentanil, and cisatracurium. Some patients received dexamethasone 5 to 10 mg or methylprednisolone 5 mg/kg, depending on the anesthesiologist's experience and preference. After anesthesia induction, 40 to 60 mL of 0.375% ropivacaine solution mixed with 1 μg/kg dexmedetomidine was used for pectoral muscle fascial plane block, involving deep serratus anterior plane block either alone or combined with pectoral nerve block I (PECS I) and pectoral nerve block II (PECS II). Anesthesia maintenance was achieved with the continuous infusion of propofol and remifentanil.

For patients undergoing UFTCA, the endotracheal tube was removed within 1 hour postoperatively when extubation criteria were met, followed by transfer to the ICU. In the CGA group, anesthesia maintenance drugs were continued until transfer to the ICU. More details are provided in Figure 1.

Data Collection and Statistical Analysis

Statistical analysis was conducted using SPSS version 26.0 (IBM). Normally distributed continuous data are presented as mean ± standard deviation and compared using repeated-measures analysis of variance. Skewed continuous data are presented as median (interquartile range) and compared using the Mann-Whitney *U* test. Categorical and ordinal data are presented as frequency (percentage) and compared using the χ^2 test or Fisher exact test, with $p < .05$ indicating statistical significance. Factors associated with UFTCA identified through univariate analysis ($p < 0.05$) were included in the multivariable logistic regression, with $p < 0.05$ considered statistically significant. Before the univariate analyses, a collinearity analysis was performed for all included variables.

Results

General Findings

The study cohort comprised 967 patients who underwent MICS at the study center between January 2021 and July 2023. Twenty patients were excluded from the analysis, including 5 patients with cardiogenic shock, 5 with septic shock, 2 requiring ECMO support, 3 with LVAD implantation, and 5 with incomplete clinical data. Finally, 947 patients were enrolled, of whom 439 (46%) underwent UFTCA and 508 (54%) underwent CGA (Fig 2). Univariate analysis of preoperative (Table 1) and intraoperative characteristics (Table 2) of the 2 groups revealed statistically significant differences in some variables.

Analysis of Factors Influencing UFTCA

In the collinearity analysis, all variables included had a tolerance coefficient <0.1 and a variance inflation factor <5, indicating that the collinearity among variables was within an acceptable range and that the current independent variables could continue to be used in the analysis (Appendix A, Table S1). Variables identified with $p < 0.05$ in univariate analysis were incorporated into a multivariate logistic regression analysis to identify the independent factors of UFTCA implementation (Table 3). The analysis revealed the following independent negative factors for UFTCA: age >50 years, a history of stroke (OR, 2.290; $p = 0.009$), higher intraoperative sufentanil dose (OR, 1.035; $p < 0.001$), CPB duration (OR, 1.013; $p < 0.001$), and surgery ending after 8 pm (OR, 2.184; $p < 0.001$). Conversely, pectoral muscle fascial plane block (OR, 0.120; $p < 0.001$) and intraoperative dexamethasone use (OR, 0.438; $p < 0.001$) were positive factors for UFTCA. The regression model exhibited good calibration (Hosmer-Lemeshow test, $p = 0.713$) and discrimination (C statistic = 0.807).

Comparison of Postoperative Recovery between UFTCA and CGA

Postoperative Basic Conditions and Complications

Patients in the UFTCA group had a significantly shorter ICU length of stay (22.83 [20] hours vs 44 [43] hours; $p < 0.001$) and

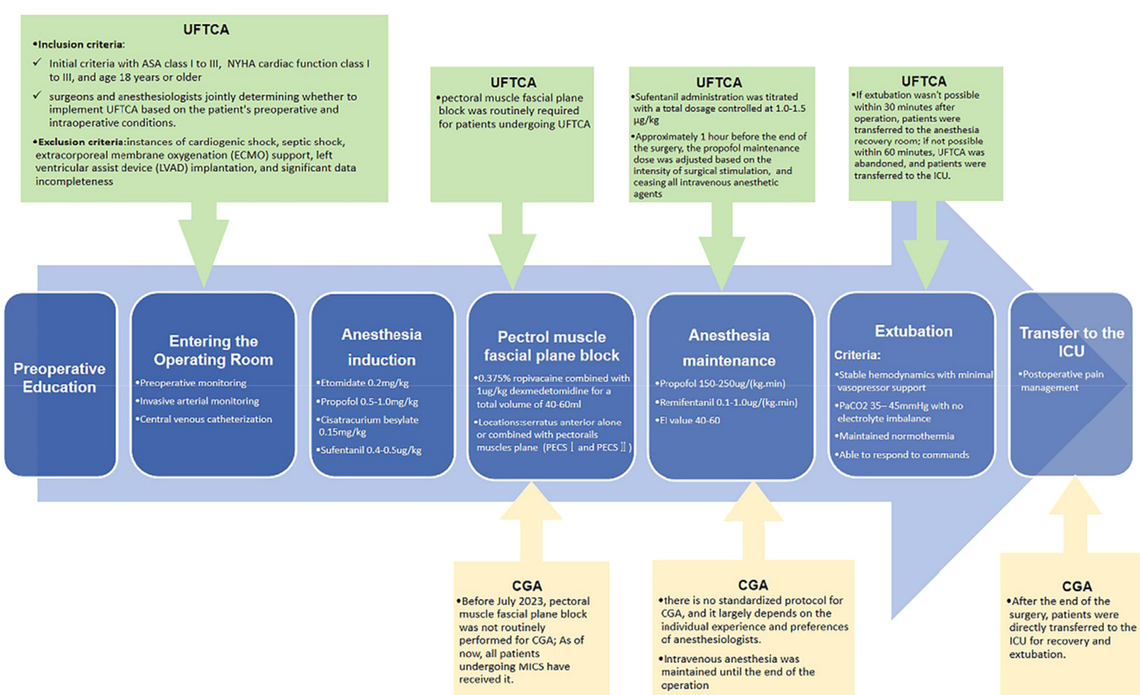

Fig 1. Anesthesia protocol in minimally invasive cardiac surgery (MICS). CGA, conventional general anesthesia; ECMO, extracorporeal membrane oxygenation; ICU, intensive care unit; LVAD, left ventricular assist device; MICS, minimally invasive cardiac surgery; PECS, pectoral muscles plane; UFTCA, ultra–fast-track cardiac anesthesia.

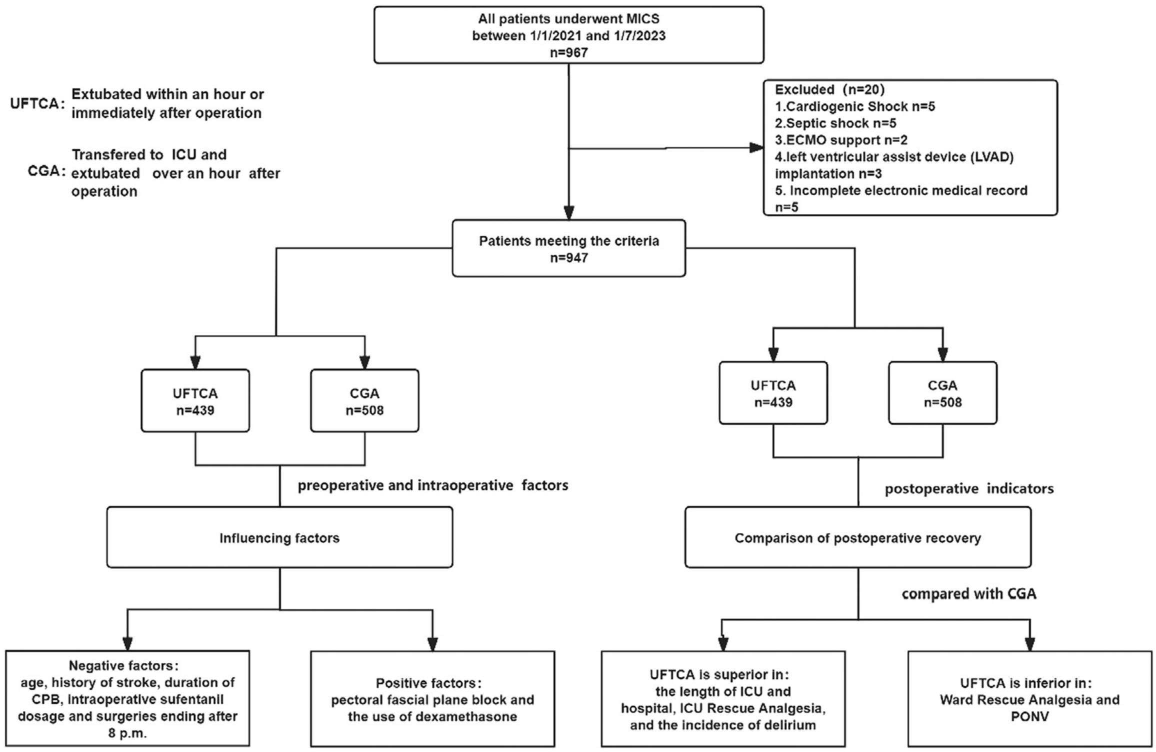

Fig 2. Flow diagram of the study population characteristics. CGA, conventional general anesthesia; ECMO, extracorporeal membrane oxygenation; ICU, intensive care unit; MICS, minimally invasive cardiac surgery; UFTCA, ultra−fast-track cardiac anesthesia

postoperative hospital length of stay (8 [3] days vs 10 [5] days; $p < 0.001$) compared to those in the CGA group (Table 4). The incidence of postoperative delirium was notably lower in the UFTCA group (6 [1.4%] vs 28 [5.5%]; $p = 0.001$). Although fewer patients in the UFTCA group required ICU rescue analgesia (77 [17.5%] vs 178 [35%]; $p < 0.001$), more patients required ward rescue analgesia (185 [42.1%] vs 158 [31.1%]; $p < 0.001$). Consequently, there was no significant overall difference between the 2 groups in the number of patients requiring postoperative rescue analgesia (223 [50.8%] for UTFCA vs 260 [51.2%] for CGA; $p = 0.906$). The UFTCA group had a higher incidence of postoperative nausea and vomiting (PONV) (86 [19.6%] vs 62 [12.2%]; $p = 0.002$). Notably, there were no significant differences between the 2 groups in the rates of postoperative stroke, atelectasis, pulmonary edema, pleural effusion, and acute respiratory distress syndrome (ARDS). The diagnosis of these complications was based on imaging evidence and medical records.

Postoperative Pulmonary Ventilation and Oxygenation Function

Both groups showed a decrease in the Oxygenation Index (OI) (OI = PaO_2/fraction of inspired oxygen [FiO_2]) at 1 hour (T1), 6 hours (T2), 12 hours (T3), and 18 hours (T4) postsurgery compared to preoperative baseline (T0), with a gradual increase from T1 to T4. The UFTCA group had statistically significant lower OI at T1 and higher OI at T2 compared to the CGA group ($p < 0.05$) (Fig 3, A; Table 5).

Both groups demonstrated an increase in the Respiratory Index (RI) (RI = $P(A\text{-}a)O_2/PaO_2$) at T1, T2, T3, and T4 compared to T0, showing a decreasing trend from T1 to T4 ($p < 0.05$), with no statistically significant differences between the 2 groups ($p > 0.05$) (Fig 3, B; Table 5).

Discussion

UFTCA strategies optimize anesthesia management and minimize opioid use, enabling extubation within 1 hour or even immediately after the operation, which aims to promote rapid postoperative recovery and efficient utilization of medical resources.[8,9] This retrospective study aimed to analyze the factors influencing the implementation of UFTCA from the perspectives of preoperative and intraoperative conditions and to compare its postoperative recovery with that of CGA.

Since 2021, the study center has used UFTCA for MICS, and it was successfully implemented in 439 cases by July 2023. This retrospective analysis identified that age >50 years, history of stroke, prolonged CPB duration, increased intraoperative sufentanil use, and surgeries ending after 8 pm were factors hindering UFTCA implementation, while pectoral muscle fascial plane block and intraoperative dexamethasone use were positive factors for UFTCA implementation. Previous research identified several risk factors associated with delayed extubation following cardiac surgery: increased BMI, older age, higher fentanyl dosage, and prolonged operation and CPB times were negatively associated with early extubation. Relatively simple surgeries, such as

Table 1
Perioperative characteristics of the study population

Characteristic	UFTCA group (N = 439)	CGA group (N = 508)	p value
Male sex, n (%)	223 (50.8)	274 (53.9)	0.335
Age, y, mean ± SD	55 ± 23	61 ± 16	<0.001
BMI, kg/m^2, mean ± SD	23.23 ± 4.16	22.84 ± 4.59	0.279
NYHA grade, n (%)			<0.001
Ⅱ	213 (48.5)	159 (31.3)	
Ⅲ	193 (44)	299 (58.9)	
Ⅳ	33 (7.5)	50 (9.8)	
ASA grade, n (%)			<0.001
Ⅱ	7 (1.6)	18 (3.5)	
Ⅲ	347 (79)	288 (56.7)	
Ⅳ	85 (19.4)	202 (39.8)	
Extubation time, min, median (IQR)	14 (13)	867.5 (662.5)	<0.001
EF <50%, n (%)	27 (6.2)	59 (11.6)	0.004
COPD, n (%)	4 (0.9)	16 (3.1)	0.031
Asthma, n (%)	2 (0.5)	1 (0.2)	0.899
Pulmonary inflammation, n (%)	14 (3.2)	38 (7.5)	0.004
Pulmonary surgery, n (%)	5 (1.1)	5 (1.0)	0.816
Myocardial infarction, n (%)	0 (0)	2 (0.4)	0.544
Congenital heart disease, n (%)	37 (8.4)	45 (8.9)	0.814
Cardiovascular surgery history, n (%)	38 (8.7)	69 (13.6)	0.017
Coronary artery disease, n (%)	39 (8.9)	82 (16.1)	0.001
Atrial fibrillation, n (%)	79 (18)	196 (38.6)	<0.001
Hypertension, n (%)	132 (30.1)	182 (35.8)	0.061
Pulmonary hypertension, n (%)	132 (30.1)	209 (41.1)	<0.001
History of stroke, n (%)	32 (7.3)	66 (13.0)	0.005
Diabetes, n (%)	22 (5.0)	49 (9.6)	0.007
Smoking history, n (%)	79(18.0%)	106(20.9%)	0.267
Drinking history, n (%)	52(11.8%)	66(13.0%)	0.594
ALT, U/L, mean SD	17 (14)	19 (16)	0.002
Creatinine, μmol/L, mean SD	71 (24)	75.95 (22)	<0.001

Abbreviations: ALT, Alanine aminotransferase; ASA, American Society of Anesthesiologists; BMI, body mass index; CGA, conventional general anesthesia; COPD, chronic obstructive pulmonary disease; EF, ejection fraction; IQR, interquartile range; NYHA, New York Heart Association; UFTCA, ultra−fast-track cardiac anesthesia.

isolated valve procedures, were positively associated with early extubation.[14−16] These results may reflect 2 aspects. First, these factors influenced clinical decision making, making anesthesiologists more inclined to use CGA, and second, these factors may affect the success of UFTCA. However, which aspect had the dominant role remains unclear, and further stratified analysis is needed. These results offer guidance for anesthesiologists contemplating UFTCA, though the final decision should be informed by the clinician's expertise.

In this study, patients in the UFTCA group were younger than those in the CGA group (Table 1). Further multivariate analysis identified age >50 years as a factor hindering UFTCA implementation at the study center (Table 3). Elderly patients may be at increased risk of postoperative complications owing to decreased pulmonary reserve, increased comorbidities, and susceptibility to tissue damage. Anesthesiologists should weigh the potential risks of complications more when deciding on whether to implement UFTCA.

The UFTCA group had significantly fewer patients with a history of stroke compared to the CGA group (Table 1). Notably, a history of stroke was identified as an independent risk factor for nonimplementation of UFTCA at the study center (Table 3). Research has suggested that a history of cerebrovascular disease (including stroke), referring to preexisting conditions affecting the brain's blood vessels or carotid artery stenosis, is a strong preoperative predictor of stroke.[17] Nevertheless, there was no significant difference in the incidence of postoperative cerebral infarction between the 2 groups (Table 4).

With the nonphysiologic circulation used during cardiac surgery, CPB can lead to pulmonary endothelial damage, and increased pulmonary vascular resistance.[18] The results of this study align with previous research[15] identifying prolonged CPB duration as an independent influencing factor for the implementation of UFTCA.

Previous studies overlooked a critical influencing factor—the completion time of surgery. Studies showed that surgeries performed at night (with induction of anesthesia between 8 pm and 7:59 am) increased the risk of intraoperative and postoperative complications.[19,20] However, surgery ending after 8 pm, including those beginning in the afternoon, was considered a critical timepoint at the study center based on the daily arrangement. Consequently, surgery ending after 8 pm was

Table 2
Intraoperative characteristics of the study population

Characteristic	UFTCA group (N = 439)	CGA group (N = 508)	p value
Type of surgery, n (%)			0.095
LVOT obstruction surgery	84 (19.1)	63 (12.4)	
Isolated MVR	118 (26.9)	145 (28.5)	
Isolated TVR	9 (2.1)	16 (3.1)	
Isolated AVR	60 (13.7)	69 (13.6)	
DVR	97 (22.1)	122 (24.0)	
Multiple valve surgery	18 (4.1)	14 (2.8)	
Atrial septal defect repair	16 (3.6)	31 (6.1)	
Aortic replacement	15 (3.4)	15 (3.0)	
Aortic valve + ascending aortic replacement	4 (0.9)	10 (2.0)	
Other	18 (4.1)	23 (4.5)	
Duration of surgery, min, median (IQR)	270 (75)	300 (106)	<0.001
Duration of CPB, min, median (IQR)	140 (51)	169 (75)	<0.001
Aortic clamp time, min, median (IQR)	91 (46)	111 (64)	<0.001
Sufentanil dosage, μg, median (IQR)	60 (20)	100 (90)	<0.001
Remifentanil dosage, μg, median (IQR)	2.7 (1)	2.0 (1)	<0.001
Methylprednisolone use, n (%)	201 (45.7)	192 (37.7)	0.006
Dexamethasone use, n (%)	377 (74.2)	131 (25.8)	<0.001
Pectoral muscle fascial plane block, n (%)	406 (92.5)	304 (59.8)	<0.001
Operations completed after 8 pm, n (%)	72 (16.4)	169 (33.3)	<0.001
Blood transfusion, n (%)	93 (21.2)	140 (27.6)	0.023

Abbreviations: AVR, Aortic valve replacement; CGA, conventional general anesthesia; CPB, cardiopulmonary bypass; DVR, double-valve replacement; IQR, interquartile range; LVOT, left ventricular outflow tract; MVR, mitral valve replacement; TVR, tricuspid valve replacement; UFTCA, ultra−fast-track cardiac anesthesia.

one of the variables in this study. The results show that surgery ending after 8 pm was inversely correlated with UFTCA implementation. The possible reason was that ending surgeries after 8 pm at the center usually indicated an overwhelming workload for the surgical team.

A high opioid strategy in traditional cardiac anesthesia aims to mitigate the stress response but might result in side effects, such as respiratory depression and delayed extubation. In this retrospective study, the dosage of sufentanil used was significantly lower in the UFTCA group compared to the CGA group (60 [20] μg vs 100 [90] μg). Consequently, the increased sufentanil dosage was inversely correlated with UFTCA implementation, aligning with the findings of Kathirvel.[15] With the use of minimal dose opioids, postoperative pain management has become a very important focus.

Table 3
Multivariate regression analysis of independent factors influencing UFTCA

Factor	B	SE	Wald	OR	95% CI	p value
Age			20.307			0.003
<40, y	Reference					
41-50	0.291	0.367	0.628	1.338	0.651-2.748	0.538
51-60	0.655	0.318	4.232	1.924	1.031-3.590	0.040
61-70	1.086	0.315	11.923	2.963	1.599-5.489	0.001
>70	1.360	0.379	12.879	3.897	1.854-8.190	<0.001
History of stroke	0.828	0.317	6.831	2.290	1.230-4.262	0.009
ALT	0.006	0.003	3.109	1.006	0.999-1.012	0.078
Creatinine	0.005	0.003	3.467	1.005	1.000-1.011	0.059
ASA grade			28.820			<0.001
II	Reference					
III	−1.019	0.636	2.572	0.361	0.104-1.254	0.109
IV	0.094	0.649	0.021	1.099	0.308-3.917	0.885
CPB duration	0.013	0.002	36.919	1.013	1.009-1.018	<0.001
Sufentanil dosage	0.034	0.004	80.251	1.035	1.027-1.042	<0.001
Dexamethasone	−0.826	0.199	17.195	0.438	0.296-0.647	<0.001
Pectoral muscle fascial plane block	−2.123	0.276	59.146	0.120	0.070-0.206	<0.001
Operations completed after 8 pm	0.781	0.219	12.760	2.184	1.423-3.354	<0.001

Abbreviations: ALT, Alanine transaminase; ASA, American Society of Anesthesiologists; CPB, cardiopulmonary bypass; CI, confidence interval.

Table 4
Postoperative general conditions and complications

Postoperative indicators	UFTCA group (N = 439)	CGA group (N = 508)	p value
Length of stay in ICU, h, median (IQR)	22.83 (20)	44 (43)	<0.001
Length of stay in hospital, d, median (IQR)	8 (3)	10 (5)	<0.001
ICU rescue analgesia, n (%)	77 (17.5)	178 (35)	<0.001
Ward rescue analgesia, n (%)	185 (42.1)	158 (31.1)	<0.001
Total postoperative rescue analgesia, n (%)	223 (50.8)	260 (51.2)	0.906
Delirium, n (%)	6 (1.4)	28 (5.5)	0.001
Stroke, n (%)	3 (0.7)	9 (1.8)	0.156
PONV, n (%)	86 (19.6)	62 (12.2)	0.002
Atelectasis, n (%)	133 (30.3)	154 (30.3)	0.995
Pulmonary edema, n (%)	16 (3.7)	27 (5.3)	0.221
Pleural effusion, n (%)	394 (89.7)	450 (88.6)	0.565
ARDS, n (%)	4 (0.9)	9 (1.8)	0.256

Abbreviations: ARDS, Acute respiratory distress syndrome; CGA, conventional general anesthesia; ICU, intensive care unit; PONV, postoperative nausea and vomiting; UFTCA, ultra−fast-track cardiac anesthesia.

Previous literature highlights that effective pain management was crucial for postoperative extubation and essential for UFTCA implementation.[21] The present results show that even with a smaller sufentanil dosage in the UFTCA group, the requirement for rescue analgesia in the ICU was significantly less than that in the CGA group. Meanwhile, more patients in the UFTCA group received pectoral muscle fascial plane blocks. This implies that pectoral muscle fascial plane blocks contributed to effective analgesia during the postoperative ICU stay.

Previous research has demonstrated the efficacy of PECS I, PECS II, and serratus anterior plane blocks in alleviating pain in cardiac and thoracic surgeries.[22−24] The present study supports the idea that pectoral muscle fascial plane block serves as a protective factor in UFTCA implementation, contributing effective analgesia with a smaller sufentanil dosage. Patients in the CGA group received more sufentanil but fewer blocks intraoperatively, and subsequently, more analgesia was required during their ICU stay.

In contrast to the lower analgesic requirements during their ICU stay, more patients in the UFTCA group required rescue analgesia compared to the CGA group after being transferred back to the ward. Research has shown that ropivacaine combined with dexmedetomidine extended the initial rescue analgesia within 24 hours postoperatively.[25] The blocking solution used was a mixture of 0.375% ropivacaine combined with 1 μg/kg dexmedetomidine. The increasing need for rescue analgesia over time might be attributed to the diminishing effect of the block, as reflected in the increased demand for rescue analgesia in the ward. This highlights the need to maintain effective pain management for patients who received pectoral muscle fascial plane blocks.

The study identified the intraoperative administration of dexamethasone as a facilitating factor for the implementation of UFTCA (Table 3). Commonly used for its anti-inflammatory effects, dexamethasone significantly reduces lung inflammation by modulating the activity of inflammatory mediators and reducing the activation and aggregation of neutrophils.[26] Research indicates that the use of dexamethasone is significantly associated with improved respiratory function after surgery.[27] At the study center, anesthesiologists may choose dexamethasone based on its anti-inflammatory properties or its role in preventing postoperative nausea and vomiting.

This research compared postoperative recovery, lung function, and complications, confirming UFTCA's importance in speeding up patient recovery after surgery. Patients identified as UFTCA had significantly shorter ICU and hospital stays, which was associated with accelerated patient recovery, optimized medical resource utilization, and reduced healthcare costs.[7,12]

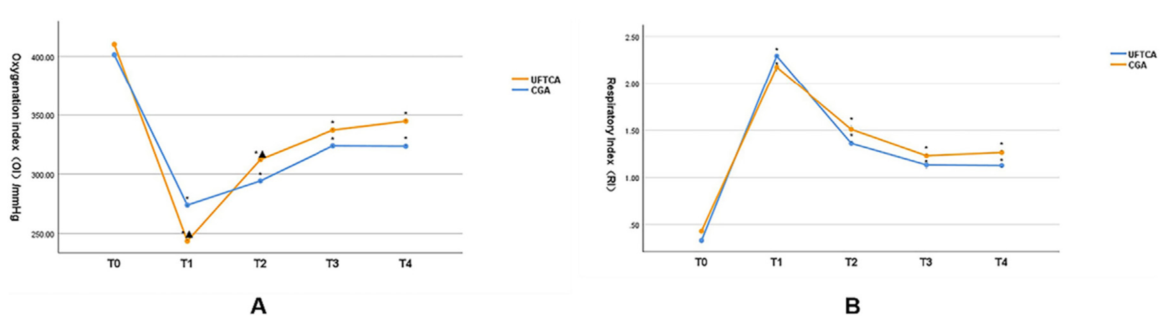

Fig 3. Comparison of the oxygenation index (OI) (A) and respiratory index (RI) (B) preoperatively and at various time points after extubation: T0, preoperative; T1, 1 hour after surgery; T2, 6 hours after surgery; T3, 12 hours after surgery; T4, 16 hours after surgery. *p < 0.05 compared to T0; ▲p < 0.05 compared to the CGA group. CGA, conventional general anesthesia; UFTCA, ultra−fast-track cardiac anesthesia.

Table 5
Trends in OI and RI

Indicator	Group	T0	T1	T2	T3	T4
OI/mmHg	UFTCA (N = 439)	410.17 ± 88.73	243.39 ± 115.12*▲	312.24 ± 116.25*▲	337.24 ± 120.11*	344.80 ± 297.33*
	CGA (N = 508)	401.34 ± 109.59	273.79 ± 123.84*	294.25 ± 104.58*	323.91 ± 106.94*	323.55 ± 126.15*
RI	UFTCA (N = 439)	0.33 ± 0.32	1.87 ± 1.34*	1.14 ± 0.82*	1.04 ± 0.86*	0.93 ± 1.25*
	CGA (N = 508)	0.45 ± 0.65	2.80 ± 1.71*	1.98 ± 1.13*	1.73 ± 1.03*	1.97 ± 1.95*

Data are presented as mean ± standard deviation.
Abbreviations: CGA, Conventional general anesthesia; OI, oxygenation index; RI, respiratory index; UFTCA, ultra−fast-track cardiac anesthesia.
*$p < 0.05$ compared to T0; ▲$p < 0.05$ compared to the CGA group.

In assessing UFTCA's impact on pulmonary recovery, we observed a temporary drop in OI levels and a rise in RI levels in UFTCA patients after extubation, with subsequent rapid improvement (Fig 2). This suggests that the temporary downturn in lung function did not impede long-term recovery. The rapid recovery of lung function observed in this study may be attributed to early extubation, which could facilitate early mobilization.

Moreover, the incidence of postoperative delirium was lower in the UFTCA group. Research indicates that longer duration of postoperative mechanical ventilation is associated with a higher incidence of delirium.[28] Notably, there were no significant between-group differences in the incidence of postoperative atelectasis, pulmonary edema, pleural effusion, or ARDS.

Although the UFTCA group used more dexamethasone, known for its antiemetic effects, their incidence of PONV incidence was higher than that in the CGA group, contrary to Kogan's earlier findings.[29] PONV is caused by multiple factors acting on multiple sites in the center and periphery through multiple pathways. The mechanisms underlying the increase PONV incidence in UFTCA patients warrant further elucidation.

The results of the present study reinforce the feasibility and benefits of UFTCA in MICS. Nonetheless, successfully implementing UFTCA still faces challenges, particularly in postoperative analgesia and PONV prevention.

Limitations

This study's retrospective observational nature could limit the universality of the results and affect their generalizability. During this retrospective observation period, the anesthesia protocols and medication regimens were not completely standardized among all anesthesiologists. The drugs and doses of rescue analgesia used by ICU physicians and ward surgeons also were not standardized, making it difficult for us to assess the differences in opioid dosing between the groups. Moreover, the 8 pm threshold was based on the study center's daily arrangement, and different centers might have alternative patterns.

Conclusion

In the present study, such factors as age >50 years, a history of stroke, prolonged duration of CPB, higher intraoperative sufentanil dosage, and surgery ending after 8 pm were identified as negative factors for UFTCA implementation. Conversely, the use of pectoral fascial plane blocks and administration of intraoperative dexamethasone were positive factors. UFTCA in MICS accelerates postoperative recovery, as reflected in shorter ICU and hospital stays, associated with a reduced incidence of delirium, and improved postoperative pulmonary function. Nonetheless, challenges remain in optimizing postoperative analgesia and preventing PONV within the UFTCA framework.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Tian Jiang: Writing − original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Linting Xu:** Validation, Software, Investigation, Data curation. **Qinghui Zheng:** Writing − review & editing, Project administration, Methodology, Investigation. **Yihui Zhang:** Writing − review & editing, Validation, Project administration, Methodology, Investigation, Data curation. **Shuaibing Wang:** Writing − review & editing, Methodology, Formal analysis, Data curation. **Yu Wang:** Writing − review & editing, Visualization, Supervision, Software, Project administration, Funding acquisition. **Xiaokan Lou:** Writing − review & editing, Supervision, Resources, Methodology, Funding acquisition. **Meijuan Yan:** Writing − review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Hanwei Wei:** Writing − review & editing, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Data Sharing Statement

The data underlying this article will be shared on reasonable request to the corresponding author.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1053/j.jvca.2025.04.009.

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